

Q.N.1 How many zeros are there at the end of $100!$

$$\Rightarrow 100! = 100 \times 99 \times \dots \times 1$$

Logic: Multiply by 10 gives zero.

$$1-10 \Rightarrow 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

Now,

$$1-10 \Rightarrow (5 \times 2), 10 \Rightarrow 2 \text{ zero}$$

$$11-20 \Rightarrow 15, 20 \Rightarrow 2$$

$$21-30 \Rightarrow 25, 30 \Rightarrow 3 \text{ zero}$$

$$31-40 \Rightarrow 35, 40 \Rightarrow 2$$

$$41-50 \Rightarrow 45, 50 \Rightarrow 3 \text{ zero}$$

$$51-60 \Rightarrow 55, 60 \Rightarrow 2 "$$

$$61-70 \Rightarrow 65, 70 \Rightarrow 2 "$$

$$71-80 \Rightarrow 75, 80 \Rightarrow 3 \text{ zero}$$

$$81-90 \Rightarrow 85, 90 \Rightarrow 2 "$$

$$91-100 \Rightarrow 95, 100 \Rightarrow 3 \text{ zero}$$

OR

$$\frac{100}{5} = 20$$

$$\frac{20}{5} = 4$$

$$24 //$$

$$\left[\begin{array}{l} (5 \times 5) \\ 25 \times 3 \end{array} \right]$$

$$24$$

Q.N.2 6 student, each student shakes hand with each other. How many hand shakes in total.

$$2 \text{ student} \rightarrow 1 \text{ handshake}$$

$$3 \quad " \quad \Rightarrow 2 \quad "$$

$$4 \quad " \quad \Rightarrow 3 \quad "$$

$$5 \quad " \quad \Rightarrow 4 \quad "$$

$$6 \quad " \quad \Rightarrow 5 \quad "$$

$$15 //$$

* Sum of first k positive integer.

$$1 + K + 1 + K = 9(n)$$

$$S(k+1) = 1 + 2 + 3 + \dots + k + k + 1$$

$$A - I - S = 2.1 + 1.$$

$$0-4=5 \Rightarrow 0.2+1$$

$$16 - 9 = 7 \quad a) 2.3 + 1$$

$$(k+1)^2 - k^2 = k^2 + 2k + 1 - k^2 = 2k + 1$$

$$(1^2 + 3^2) + (3^2 + 5^2) + (5^2 + 7^2) + \dots + (k+1)^2 - k^2 = (k+1)^2 - 1^2$$

$$\Rightarrow \begin{aligned} & 2 \cdot 1 + 1 \\ & + 2 \cdot 2 + 1 \\ & + 2 \cdot 3 + 1 \\ & + 2 \cdot k + 1 \end{aligned}$$

$$\Rightarrow 2(1+2+3+\dots+k) + (1+1+1+\dots)$$

$$\text{Ans. } (x^2 - 1)^2 = 25(x) + K$$

$$f(x) + g(x) + h(x) = 25(x) + x$$

$$u(x) = x^2 + x$$

$$3(x) = x(x+1)$$

$$x^2 + 7x + 12 = (x+4)(x+3) = 0$$

$$S = (1+2+3+\dots+(n-1)) - (2+3+4+\dots+(n-2))$$

$$f(x) = \frac{1}{2} (1 + x + x^2 + \dots + x^{n-1})$$

$$p(x) + q(x) = 2 \left((x-1)^2 + (x-1) \right)$$

✓ $(x-1)(x+1) = (x-1)(x+1)$

$$1) \quad 25 - 21 - 10 = 2 \quad \frac{-2k^2}{2} = -k^2 //$$

* Sum of ^{square} ~~cube~~ of k Positive integer.

$$1^2 + 2^2 + 3^2 + 4^2 + \dots + k^2 \Rightarrow T(k).$$

$$r^3 - (r-1)^3 = 3r^2 + 3r + 1$$

Now, $2^3 - 1^3 = 8 - 1 = 7 = 3 \cdot 1^2 + 3 \cdot 1 + 1$

$$3^3 - 2^3 = 27 - 8 = 19 = 3 \cdot 2^2 + 3 \cdot 2 + 1$$

$$4^3 - 3^3 = 64 - 27 = 37 = 3 \cdot 3^2 + 3 \cdot 3 + 1$$

$$(k+1)^3 - k^3 = k^3 + 3k^2 + 3k + 1 - k^3 = 3k^2 + 3k + 1$$

Now $2^3 - 1^3 + 3^3 - 2^3 + 4^3 - 3^3 + \dots + (k+1)^3 - k^3 \Rightarrow (k+1)^3 - 1^3$

L.H.S. $3(1^2 + 2^2 + 3^2 + \dots + k^2) + 3(1 + 2 + 3 + \dots + k) + (1+1+\dots+1)$
 $3 \cdot T(k) + 3 \cdot \frac{(k^2 + k)}{2} + k$

$$(k+1)^3 - 1^3 = 3T(k) + 3 \frac{(k^2 + k)}{2} + k$$

~~$T(k) =$~~

$$k^3 + 3k^2 + 3k + 1 - 1 = 3T(k) + 3 \frac{(k^2 + k)}{2} + k$$

$$3T(k) = k^3 + 3k^2 + 3k - \frac{(3k^2 + 3k)}{2} - k$$

$$= 2k^3 + 6k^2 + 6k - 3k^2 - 3k - 2k$$

$$= \frac{2k^3 + 9k^2 + 11k}{2}$$

$$= \frac{k(2k^2 + 9k + 11)}{2}$$

$$= \frac{2k^3 + 9k^2 + 11k}{2}$$

$$= \frac{k(2k^2 + 9k + 11)}{2}$$

$$= \frac{k}{6} (2k^2 + 9k + 11)$$

$$= \frac{k}{6} (k+1)(2k+1)$$

* K-Student:- Every student shakes hand with each other student - if K is even, are the total handshakes even or odd.

0	0	even
1	0	even
2	1	odd
3 (1+2)	3	odd
4 (1+3)	6	even
5	10	even
6	15	odd
7	21	odd
8	28	even
9	36	even

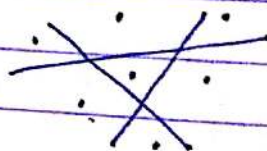
$L = 0, 1, 2, 3, 4, 5, \dots$
 $4L, 4L+1$ is even.
 $4L+2, 4L+3$ is odd. //

* With 3 straight lines, what is the max^m no. of ways to divide a 2-D plane.

1-line : 2 regions $\rightarrow$

2-line : 4 regions $\rightarrow$

3-line : 7 regions $\rightarrow$



* What is the last digit of 7^{65432} ?

$$7^1 = 7$$

$$7^2 = 49$$

$$7^3 = 343$$

$$7^4 = 2401$$

$$7^5 = 16807$$

$$7^6 = 117649$$

$$7^7 = 823543$$

$$7^8 = 5764801$$

$$8 \overline{) 65432} \quad (8179$$

$$\therefore (7^8) 8179 \cdot 7^0$$

$$= 1 \cdot 1$$

ends with 1 //

last digit of 3^{4798}

$$3^1 = 3$$

$$3^5 = 243$$

$$3^2 = 9$$

$$3^6 = 729$$

$$3^3 = 27$$

$$3^7 = 2187$$

$$3^4 = 81$$

$$3^8 = 6561$$

Here, $4 \mid 4798 \Rightarrow 1199 \cdot 4$

$$\begin{array}{r} 4 \\ 07 \\ 4 \\ \hline 39 \\ 36 \\ \hline 38 \\ -36 \\ \hline 2 \end{array}$$

$$(3^4)^{1199} \cdot 3^2$$

$$= 1 \times 9$$

$$= 9$$

$\therefore$ so last digit is 9.

How many zeros end

$$2^{300} \cdot 5^{600} \cdot 4^{400}$$

$$\Rightarrow 2^{300} \cdot 5^{600} \cdot 2^{2(400)}$$

$$\Rightarrow 2^{1100} \cdot 5^{600}$$

$$= 2^{600} \cdot 2^{500} \cdot 5^{600}$$

$$= 2^{500} \cdot (2 \times 5)^{600}$$

$$= 2^{500} \cdot 10^{600}$$

$$= 10^{600} \Rightarrow 600 \text{ zeros.} //$$

* We are given K objects $\{a_1, \dots, a_K\}$. How many different ordered pairs may be made up of those K objects?

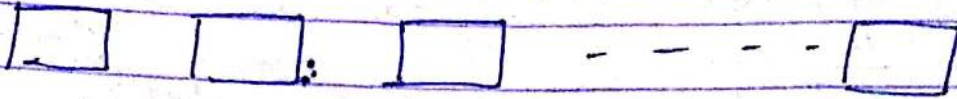
Solⁿ: There are K possible choices (namely $a_1, a_2, \dots, a_K$) for the first element of ordered pair.

After that, for next element we will have $(K-1)$ choices. i.e. if we choose a_1 as a first object, we may choose $a_2, a_3, \dots, a_K$ for second element. i.e. $(K-1)$ choices.

$$\text{i.e. } K \cdot (K-1).$$

* We are given K objects $\{a_1, \dots, a_K\}$. In how many diff. orders can we arrange these objects?

Solⁿ: Suppose that we have K positions into which ~~to~~ put the objects. There are K objects $(a_1, a_2, \dots, a_K)$ that we may put in the first position.



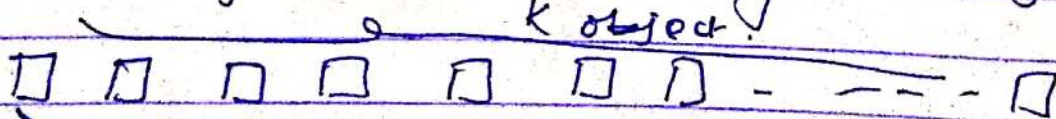
K position

Having placed an object into the first position, there remain $(K-1)$ different objects to put into the second position. i.e. $K \cdot (K-1)$ choices of pairs of objects to put into the first two positions.

1 st	K
2 nd	K-1
3 rd	K-2
⋮	⋮
⋮	3
⋮	⋮
⋮	1

$$\text{i.e. } K(K-1)(K-2) \dots 3 \cdot 2 \cdot 1 \Rightarrow K! //$$

* We are given K objects $\{a_1, a_2, \dots, a_K\}$. Suppose that m is a positive integer that is less than or equal to K . In how many ways can we choose m objects from among the original K .



first m objects.

for m objects $= m!$

for $(k-m)$ objects $= (k-m)!$

$$\therefore \frac{k!}{m!(k-m)!}$$

* How many hands are possible in 5 card poker?

$$\Rightarrow \binom{k}{m} = \binom{52}{5} \therefore C = \frac{k!}{m!(k-m)!} = \frac{52!}{5!47!} = 2,598,960 //$$

* How many pairs of bridge hands may be dealt from a deck of 52 cards

$\Rightarrow$ The first team member is dealt 13 cards from among the total of 52. The no. of possible hands that this person could be dealt is

$$C_1 = \binom{52}{13} = \frac{52!}{13!39!}$$

The second team member is also dealt 13 cards, chosen from among the remaining 39 cards.

$$C_2 = \binom{39}{13} = \frac{39!}{13!26!}$$

$\therefore$ Total no. of pairs of hands for the two team members is

$$C_1 \cdot C_2 = 5.1578 \times 10^{21}$$

Principle Mathematical Induction

When a statement is true for a natural number $n=k$, then it will also be true for its successor $n=k+1$. and the statement is true for $n=1$.

i.e. $P(1)$ is true

$P(k+1)$ is true. //

* prove that the sum of the first n natural numbers is given by the formula

$$1+2+3+\dots+k = \frac{k(k+1)}{2}$$

Step 1.

$$P(1) = \frac{1(1+1)}{2} = \frac{2}{2} = 1, \text{ which is true.}$$

Step 2.

$$P(k+1) = \frac{(k+1)(k+2)}{2}$$

$$\text{Now, } 1+2+3+\dots+k+(k+1) = \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$= \frac{k(k+1) + 2(k+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2} \text{ which is also true.}$$

$$* \quad 1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

$P(1)$

$$P(k+1) = \frac{(k+1)(2k+3)(k+2)}{6}$$



* Assume that K is a positive integer. If $(K+1)$ letters are delivered to K mailboxes then show that one mailbox must contain at least two letters.

Solⁿ:

The statement $P(K)$ is "if $(K+1)$ letters are delivered to K mailboxes then some mailbox must receive at least two letters".

For the case $K=1$, notice that if $K+1=2$ letters are delivered to $K=1$ mailbox then ~~some~~ one mailbox will receive two letters, which proves $P(1)$.

Now Assume that $(j+1)+1$ letters have been delivered to $(j+1)$ mailboxes.

- If the last mailbox is empty, then all the letters have been delivered to the first j mailboxes. In particular, at least $(j+1)$ letters have been delivered to these first j mailboxes, which means one of these first j mailboxes contains at least two letters.

- If the last mailbox contains precisely one letter, then the remaining $(j+1)$ letters have been delivered to j mailboxes.

- If the last mailbox contains two letters, then we can say that last mailbox contains at least two letters.

Subset problem

Suppose that S is a set with K elements. Show that S has precisely 2^K subsets.

$$\Rightarrow \text{Set } S = \{s_1, s_2, s_3, \dots, s_K\}$$

$$\textcircled{I} \text{ Let } K = \emptyset \text{ (empty set)} \Rightarrow S_0 = 2^0 = 1 //$$

$\textcircled{II}$ For $(K+1)$:-

$$S_{K+1} = \{s_1, s_2, s_3, \dots, s_K, s_{K+1}\}$$

Here All subsets of S_K are also subsets of S_{K+1} : 2^K subsets.

If A is a subset of S_K then $\{A \cup S_{K+1}\} : 2^K$ subsets.

$$\therefore \text{Total} = 2^K + 2^K \Rightarrow 2^{K+1} //$$

verify Euler's formula for admissible graph

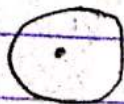
$$\boxed{V - E + F = 2}$$

$\Rightarrow$ Let $V \rightarrow$ vertices.

$E \rightarrow$ edges

$F \rightarrow$ faces.

According to the defⁿ of admissible graph, it contains single vertex and nothing else.



$$\text{i.e. } V=1, E=0, F=1.$$

$$V - E + F = 1 - 0 + 1 = 2, \text{ which is true.}$$

The next most complex graph contains one edge, with a vertex on each end & nothing else.

$$V=2, E=1, F=1.$$



$$V-E+F \Rightarrow 2-1+1 \Rightarrow 2, \text{ which is also true.}$$

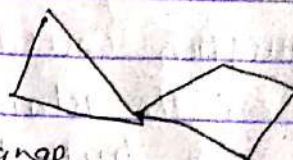
for $P(K+1) \Rightarrow$

$$V = V' + 1.$$

$$E = E' + 1.$$

$$F = F'$$

- No face change
→ vertex increased by 1
→ edge "



In example

$$V-E+F$$

$$E = 8$$

$$= V' + 1 + F' = 2$$

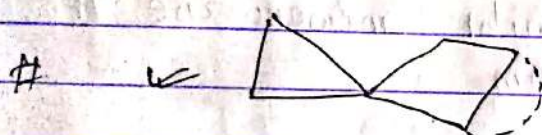
$$V = 7$$

$$\Rightarrow V' + 8 + F' = 2$$

$$F = 2$$

$$7 - 8 + 2$$

$$V-E+F = 2 //$$



(No vertex addition)

$$E = 8$$

$$V = 6$$

$$F = 4$$

$$\Rightarrow 6 - 8 + 4$$

$$\Rightarrow 2 //$$

$$V = V'$$

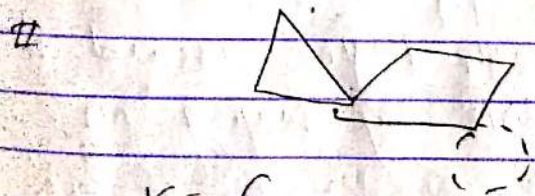
$$F' = F' + 1$$

$$E = E' + 1$$

$$V-E+F = 2$$

$$\Rightarrow V' - E' - 1 + F' + 1 = 2$$

$$\Rightarrow V' - E' + F' = 2 //$$



$$V = 6$$

$$E = 8$$

$$F = 4$$

$$\Rightarrow 6 - 8 + 4$$

$$\Rightarrow 2 //$$

$$V = V'$$

$$E = E' + 1$$

$$F' = F' + 1$$

$$V-E+F = 2$$

$$\Rightarrow V' - E' - 1 + F' + 1 = 2$$

$$\Rightarrow V' - E' + F' = 2 //$$

which follows $V-E+F=2$

Problems of Logic

* You are on an island that is populated by two types of people; truth tellers and liars. When asked a Yes-No question, a truth teller always tells the truth. And liar always lies. There is no visual method for telling a truth teller from a liar. What single question could you ask anyone that you meet on the island to determine whether that person is a truth teller or a liar?

⇒ * If we ask a direct question such as "Are you a truth teller?"

Truth teller replies "Yes"

Liar " " "Yes" (He has to lie)

* Are you a ~~truth teller~~ liar?

Truth teller replies "No"

Liar " " "No" (he has to lie)

In both case, similar answers are those so we can't separate them.

So, there must be conditional question (i.e. which contains (if-then, or, and)).

"If you are a truth-teller" then how would you answer the question "are you a liar?"

Here obviously truth teller would answer the question "are you the liar?" by saying "No".

A liar can think just as clearly as truth-teller. He also knows that a truth-teller. ~~In that case~~ if asked whether he/she is liar, will say "No". But the liar must lie, so he will say "Yes".

Which is the question that the person answer as "No" are truth-teller and "Yes" as liar?

Prove that $\sum 3^n = \frac{1}{2} (3^{n+1} - 1)$

for $n=0 \Rightarrow 3^0 = \frac{1}{2} (3^{0+1} - 1)$

$1 = \frac{2}{2} \Rightarrow 1=1$ which is true for $n=0$.

for $n=k$

$$3^0 + 3^1 + \dots + 3^k = \frac{1}{2} (3^{k+1} - 1)$$

Now, for $n=k+1$

$$3^0 + 3^1 + \dots + 3^k + 3^{k+1} = \frac{1}{2} (3^{k+1} - 1) + 3^{k+1}$$

$$\Rightarrow \frac{1 \cdot 3^{k+1}}{2} - \frac{1}{2} + 3^{k+1}$$

$$\boxed{\frac{1 \cdot 3^{k+2}}{2} - \frac{1}{2}}$$

$$\Rightarrow \frac{3 \cdot 3^{k+1}}{2} - \frac{1}{2}$$

which is true for $n=k+1 //$

* $3^n - n$ is divisible by 3 for every positive integer.

for $n=1$

$$3^1 - 1 = 2 \text{ is divisible by 3}$$

for $n=k$

$$3^k - k \text{ is divisible by 3.}$$

Now for $n=k+1$

$$p(k+1) = (k+1)^3 - (k+1)$$

$$= (k+1) ((k+1)^2 - 1)$$

$$= (k+1) (k^2 + 2k + 1 - 1)$$

$$= k^3 + 2k^2 + k^2 + 2k$$

$$= (k^3 - k) + (3k^2 + 3k)$$

$$= \underbrace{(k^3 - k)}_{\text{divisible by 3}} + 3 \underbrace{(k^2 + k)}_{\text{divisible by 3}}$$

divisible by 3 divisible by 3.

hence it is true for $n=k+1$

$5^n - n$ is divisible by 5 for every positive integer

for $n=1$: $5^1 - 1 = 0$ which is divisible by 5.

$n=K$; $(K^5 - K)$ which is divisible by 5.

$$\begin{aligned} n=K+1 & \quad (K+1)^5 - (K+1) \\ & = (K^5 + 5K^4 + 10K^3 + 10K^2 + 5K + 1) - (K+1) \\ & = (K^5 - K) + 5(K^4 + 2K^3 + 2K^2 + K) \\ & = \text{divisible by } 5 \quad \text{divisible by } 5. \end{aligned}$$

for $n=K+1$ is also true, Hence proved.

Determine how many zeros end the number $800!$.

* Multiple of 5 $\Rightarrow \frac{800}{5} = 160$.

" " 25 $\Rightarrow \frac{800}{25} = 32$

" " 125 $= \frac{800}{125} = 6$

" " 625 $= \frac{800}{625} = 1$.

$\therefore \text{Total} = 160 + 32 + 6 + 1 = 199$ //

Calculate sum. $\frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{n+1!}$

$$\Rightarrow \frac{1}{2!} + \frac{2}{3!} + \frac{3}{4!} + \dots + \frac{n}{n+1!}$$

$$\Rightarrow \frac{2-1}{2!} + \frac{3-1}{3!} + \frac{4-1}{4!} + \dots + \frac{n+1-1}{n+1!}$$

$$= 1 - \frac{1}{2!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{3!} - \dots - \frac{1}{n!} + \frac{1}{n!}$$

$$= 1 - \frac{1}{n+1!} //$$

How many zeros at the end of

$$16^{300} \times 15^{500} \cdot 250!$$

$$\Rightarrow (2^4)^{300} \times (3 \times 5)^{500} \cdot 250!$$

$$(2 \times 2 \times 5)^{50} \cdot 3^{50}$$

$$= 2^{1200} \cdot 3^{500} \times 5^{500} \cdot 250!$$

$$2^{300} \cdot 5^{150} \cdot 3^{50}$$

$$= 2^{900} \cdot 5^{350} \cdot 3^{450} \cdot 250!$$

$$= 2^{350} \cdot 2^{550} \cdot 5^{350} \cdot 3^{450} \cdot 250!$$

$$= (10)^{350} \cdot 2^{550} \cdot 5^{350} \cdot 3^{450} \cdot 250!$$

for $250!$ $\Rightarrow$ multiple of 5 = $\frac{250}{5} = 50$

" " 25 = $\frac{250}{25} = 10$

" " 125 = $\frac{250}{125} = 2$

Total = 62

$\therefore$ Total no. of zero = $350 + 62$

= 412

Suppose you are told in advance that 10 of the cattle present are lame and only have 3 feet. But Count yields 120 heads & 300 feet. How many cattle and people are there?

$$\Rightarrow \text{Total no. of heads} = 120$$

$$\quad \quad \quad \text{feet} = 300$$

Let $x \rightarrow$ no. of cattle
 $y \rightarrow$ no. of people

Then

$$x + y = 120 \quad \dots (i)$$

$$2x + y + 10 \times 3 + (x - 10) \times 4 = 300$$

Solving (i) & (ii)

$$2. (120 - x) + 30 + (x - 10) \times 4 - 40 = 300$$

$$240 + 2x - 10 = 300$$

$$\therefore x = 35$$

$$\therefore y = 120 - 35$$

$$= 85$$

* How many digits used to No. of the pages of a book having 750 pages $\rightarrow$ numbered from 1-100?

$$\Rightarrow \text{page from 1 to 9} \Rightarrow \text{total digit} = 9$$

$$\quad \quad \quad \text{"} \quad \quad \quad \text{10 to 99} \Rightarrow \quad \quad \quad = 90 \times 2 = 180$$

$$\quad \quad \quad \text{"} \quad \quad \quad \text{100 to 750} \Rightarrow \quad \quad \quad = 1953$$

$$\therefore \text{Total no. of digit} = 2142 \quad // \quad (9 + 180 + 1953)$$

A herd of Cows Invades a dangle Causing damage in the field. The boys were chasing them. There are 130 heads and 300 feet but 5 cows have 3 legs and 2 men have 1 leg. Find the no. of cows and boys.

$$\Rightarrow \therefore \text{Total no. of Head} = 130$$

$$\therefore \text{feet} = 300.$$

Let x : no. of cows

y : " " boys

$$\therefore x + y = 130 \dots (1)$$

$$5 \times 3 + (x - 5) \times 4 + 2 \times 1 + (y - 2) \times 2 = 300 \dots (2)$$

Solving (1) and (2).

$$15 + 4x - 20 + 2 + 2 \times (130 - x) - 4 = 300.$$

$$\Rightarrow 15 + 4x - 20 + 2 + 260 - 2x - 4 = 300.$$

$$\Rightarrow 2x = 47.$$

$$\therefore x = \frac{47}{2}$$

$$y = \frac{213}{2} //$$

There are more adults than boys, more boys than girls and more girls than families. If no family member has fewer than 3 children, what is the least no. of families there could be?

⇒ $A = \text{adult}, \text{Boys} = B, \text{Girls} = G, \text{families} = F$
 $A > B > G > F$

• If there were just one family, then there could be at least two girls, at least 3 boys and at least 4 - adult, which make two families which is a contradiction.

• If there were just 2 families then there would be at least 3 girls, at least 4 boys at least 5 adults. But 5 adults can be just two families which is contradiction.

• If there were just 3 families then there would be at least 4 girls, at least 5 boys and at least 6 adult. which is not a contradiction. Thus 3 families might satisfy the conditions.

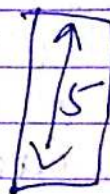
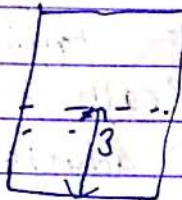
∴ Total no. of families = 3 //

We wish to bring the river 1 quart from
 of water, but we are only have 8 quart
 container and 5 quart container, How do
 we do it?

⇒ ~~Fill~~ ^{pour} 8 litre quart



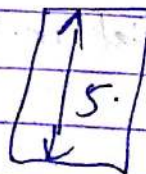
⇒ ~~Pour 5 quart~~ ~~Fill~~ pour 5 lit into
 5 quart.



⇒ Empty 5 quart and pour 3 quart
 from 8 quart container to it then
 fill up 8 quart container again and top-off
 the 5 quart container using the 8 quart
 container. 6 quarts still
 remains in 8 quart container.

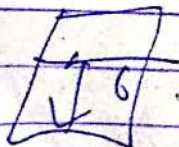


8



5 quart.

⇒ Empty 5 quart container.

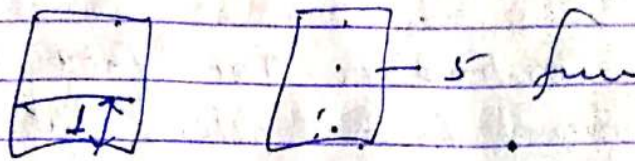


8



5

⇒ Then pour 6 quart of 8 quart container in 5 quart container then 1 quart remaining in the quart container.



* prove $\sqrt{2}$ is irrational

The rational numbers includes all positive numbers, negative numbers and zero that can be written as a ratio (fraction) of one number over another.

Proof: Let $\sqrt{2}$ is a rational number.

So $\sqrt{2} = \frac{p}{q}$ where p & q are integers $q \neq 0$ and $\text{HCF}(p, q) = 1$ that means p and q have no common factor other than 1.

$$\Rightarrow p = q\sqrt{2} \quad \text{--- (1)}$$

$$\Rightarrow p^2 = 2q^2$$

i.e. p^2 is divisible by 2.

$\therefore p$ is divisible by 2

So, 2 is a factor of p .

We shall put, $p = 2m$ in eqⁿ (1).

$$2q^2 = 4m^2$$

$$q^2 = 2m^2$$

q^2 is divisible by 2

$\therefore q$ is divisible by 2.

and 2 is a factor of 9.

$\therefore$ 2 is common factor of p & 9.

But this contradicts the fact that p & 9 have no common factor other than 1.

Thus contradiction occurs because of our assumption that $\sqrt{2}$ is rational number, which was wrong assumption. Hence, square root of 2 is an irrational number.

* Prove $\sqrt{3}$ is irrational number.

4 Explain why, if two positive real no.s sum to 100, then their product can't be 3000.

Solⁿ: Let x & y be two positive real numbers then $x+y=100 \dots (i)$

if the product of these two numbers is 3000.

i.e. $x \times y = 3000$.

$$\Rightarrow x \times (100-x) = 3000$$

$$\Rightarrow 100x - x^2 = 3000$$

$$\Rightarrow x^2 - 100x + 3000 = 0$$

$$\therefore x = \frac{-(-100) \pm \sqrt{(-100)^2 - 4 \times 1 \times 3000}}{2 \times 1}$$

$$= \frac{100 \pm \sqrt{-2000}}{2}$$

which gives imaginary number, but by question x & y are positive real numbers. Hence by contradiction, product of these two numbers never 3000.

* Explain why there are infinitely prime numbers.

Suppose there are only finitely many primes, let's say n of them. We denote them $p_1, p_2, \dots, p_n$.

Now construct a new number.

$$p = p_1 * p_2 * p_3 * \dots * p_n + 1.$$

Clearly, p is larger than any of the primes, so it doesn't equal one of them. Since $p_1, p_2, \dots, p_n$ constitute all primes p can't be prime. Thus it must be divisible by at least one of our finitely many primes, say p_i . When we divide p by p_i , we get remainder 1. That's contradiction, so our original assumption that there are finitely many primes must be false. Thus there are infinitely many prime numbers.

* A goat takes 6 days to finish grass
A ship " 3 " " " "
A cow " 2 " " " "

How much time would it take to all 3 together?

$\Rightarrow$ In 2 days 1 cow finishes.
In 1 " 1×2 " "

like In 6 days 1 goat.
In 1 " 6×1 goat.
 $\therefore$ i.e. 2 cow = 6 goat.

$\therefore$ 1 cow = 3 goat.

In 3 days 1 ship finishes

In 1 day 3 ship finishes.

$\therefore$ 3 ship = 2 goat.

1 ship = 2 goat.

1 cow + 1 ship + 1 goat

$\Rightarrow$ 3 goat + 2 goat + 1 = 6 goat

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3+2+1}{6} = \frac{6}{6} = 1$$

Here 1 goat takes 6 days
6 " " 1 days.

2) 15 teams play in a tournament.
3 pts for win, 2 pts for draw and 1 for a loss. All the team end up with 9 different no. of points, worst scored 21 pts. Show that the score of best team at least 1 game draw.

⇒ 15 teams.

$$14 + 13 + 12 + \dots + 1 = 105 \text{ game}$$

$$\text{Total pts (wins + loss)} 3 + 1 = 4 \times 105 = 420 \text{ pts.}$$

21 is the last pts. So in 14 games.

$$21 + 22 + 23 + \dots + 35 = 420 \text{ pts.}$$